

Project Guidelines – MSc Big Data Analytics

Version 1.0

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(Adopted from Robert Gordon University -MSc Project Guidelines)

1 Project CampusMoodle pages

A variety of information about projects can be found on the module CampusMoodle page. These give details of project allocations, supervisors, deadlines, guidelines, forthcoming events, and links to other useful documents. The module code is CMM799.

2 Introduction

The award of MSc requires the completion of an MSc project. It carries 60 credits and is therefore a major part of the 180-credit MSc course. This document presents guidelines for the completion of an MSc project. It is unlikely that it will cover all the difficulties a student may encounter during the project. It is extremely important, therefore, that the student **keep in regular contact with his or her supervisor** so that problems are detected and solved as soon as possible.

3 Structure of the Project Module

The MSc project requires that a student conducts a thorough investigation, formulate a project proposal, initiate the project and complete it with a prototype. A pure research based project is not valid – you must prove it via a prototype. The duration of the module is two semesters.

4 Aims of Modules

To develop a specification for a substantial problem in depth, making use of principles and practices acquired elsewhere in the course. To develop a solution from this specification through to implementation and report on the results within a fixed time frame. To enhance and develop transferable skills, communication skills and the experience of working on a major project in a guided but independent fashion.

The aim would be achieved by undertaking an investigation of a technical problem, finalizing the requirements and subsequently solving the problem. Researching into the problem domain, critically examining relevant methods and tools, gain some understanding of the problem and its domain, and develop a specification for a substantial professional (or equivalent) project relevant to the course of study is expected.

The literature review need to be focused on providing : (i) **explanation and justification** for all aspects of the problem to be tackled; (ii) a **critical review** of the approaches considered, and support for the work to be done in the project module and: (iii) a discussion on **professional, social, legal and ethical issues** related to the project. The implementation of the project should result in a software prototype that may or may not interface with a hardware.

5 Learning Outcomes for Modules

On completion of this module, students are expected to be able to:

1. Critically appraise literature and current practice in relation to a major big data project and to place it in the context of related work.

2. Plan the project, select appropriate methods, tools and technologies to carry-out research and test potential solutions that complies with the legal, ethical, social and professional context of the project.
3. Evaluate alternative design and development choices to systematically identify the best strategy to deliver a big data artifact.
4. Reflect on project findings and draw insights to establish opportunities for desirable extensions and future work.
5. Communicate effectively in a written dissertation and oral presentation to detail and defend the project contributions and research conclusions.

6 Supervision

The student would be able to select a supervisor who is within the academic panel of IIT. The student is expected to meetings with his/her supervisor. The supervisor should:

- Help the student to clarify the topics and focus the project work; direct the student to relevant areas of information (if possible)
- Recommend appropriate methodologies
- Read and comment on draft versions of parts of any written reports
- Advise the student about managing the timetable of project activities.

It is **not** the role of the supervisor to **manage the dissertation** or **the student's time** or to **motivate the student**. The student should manage his or her time, do the work and write the dissertation. The supervisor should give guidance and direction, **not do the project** for the student.

The supervisor may be away at times, especially during vacation. The supervisor should inform the student about any long absence and should discuss with the student what the student will do during that time. A second supervisor *may* be assigned to the student for that period, but this is not always possible.

7 Project Proposal

All students should produce a brief project description document which specifies the project that they have agreed to undertake. It should contain the background to the project as well as the project aims and objectives.

The student should complete the project description, with the assistance of their supervisor and submit it to the appropriate dropbox on CampusMoodle as the first project deliverable. This is required to ensure the project has sufficient depth and does not conflict with a similar topic submitted by other students. You should have already selected a supervisor when submitting the proposal.

8 Ethical Review

A student project ethical review (SPER) form must be filled in for each project. The form must be filled in even if the student and supervisor believe that there are no ethical issues. Details of the procedure, including the form to be filled in, may be found on the CampusMoodle page. The student should complete this form, with the assistance of their supervisor, and submit it to the appropriate dropbox on CampusMoodle along with the final project report.

9 Turnitin - Plagiarism detection service

The Turnitin plagiarism detection service (see www.turnitin.com) compares documents submitted by students with a variety of electronic sources, in order to identify possible occurrences of plagiarism. The service generates an annotated version of a submitted document, indicating all areas of apparent overlap with other sources, together with an overall “plagiarism index”. **The module dropboxes on CampusMoodle are linked to the Turnitin plagiarism detection service so that when you submit your work, it is also submitted to Turnitin.**

A student may submit a draft version so that they are made aware of parts of the report which appear to duplicate existing sources, and hence may give rise to an accusation of plagiarism. This will enable the student to make sure that all such occurrences are appropriately acknowledged, or else to re-write or remove them. The student may submit several revisions of the report, if necessary.

10 Duration of the Project

The time allocated to the project (both modules) is two Semesters in part time mode for a total credit of 60. The student is expected to invest 600 Hours of effort, spread thorough two semester.

11 Attendance and Participation

The entire project process lasts for several months. This can seem a long time at the start, but to ensure a satisfactory result, it is important that the student works consistently throughout both project periods. The student needs to meet regularly with his or her supervisor (normally once a week).

Module Classes

Students are advised to attend module classes once a week. The goal of attending the class is to invest at least solid three hours on working on an aspect of the module.

Attendance is recorded which is used as evidence for continuous progress for the project.

Weekly Log Sheet

Students are expected to fill the weekly log sheet, indicating the work completed and what is planned for the week. The log sheet need to be maintained (either as an electronic or manual) for each week and submitted along with the project. Meeting with the supervisor could be minuted in the log sheet.

Contact points

A contact point is defined to be an occasion when a student makes contact with a member of staff and provides evidence of participation in the course. During the project, participation will consist of concrete evidence of progress. If a student turns up for a meeting but with no evidence of progress on the project, this will not constitute a contact point.

Project

Project assessment does not include an explicit conduct mark. However, to ensure the student's regular participation in the project, project assessment requires that at least SEVEN contact points have been recorded during the period of the project and prior to the demonstration and report submission. If the criterion of SEVEN contact points during the project period is not met, then the project report will not be marked, and the project will fail, with an automatic grade F. Attendance to a module class and showing work in progress would constitute as a contact point.

12 Project Deliverables

Project Proposal

The time allowed between the start of the project and the submission of the first deliverable varies according to the cohort's overall schedule, but is normally between one and three weeks. The deadlines can be found on the CampusMoodle module pages.

MSc Project Report + Ethical Review Form + Prototype

The project report will describe the investigation, specification, design, implementation and evaluation of the software solution. The student must upload the final project report to the Turnitin drop-box on the CampusMoodle project page. The student MUST upload ALL other project materials to the appropriate drop-box on the Moodle project page. These materials are:

- Presentation slides
- Source code for the application. This should be presented in a form that will allow the markers to run the application if they wish. Where it is not possible or appropriate to provide source code, students can instead upload a video capture of a demonstration of their system.
- Slides used in the demo/presentation
- Any other materials that you consider relevant (if appropriate).

The student must submit the Ethical Review form, Two printed copies of the project report along with a CD containing the above to the school office.

In the Project Demonstration, the student will give a practical demonstration of the software solution and justify the work that has been undertaken before a panel consisting of the project supervisor and second supervisor and an independent chairperson.

13 Assessment

The student work is assessed independently by 2 markers (supervisor and second marker). The markers then meet to agree final marks. If a consensus cannot be reached, a third marker will assess the work.

The weighting of this module is 60 credits, so it is equivalent to four taught modules.

The assessment is divided into two components: project report, and demonstration. Please refer to the *Project Instructions and Grids* for an explanation of how this module is assessed.

The grading grid at the end of the above document describes the criteria used for determining a project's grade.

14 Document Specifications

The project report should include:

- a front cover (available from the School Office);
- a title page (see below);
- a signed declaration (see below) indicating that:
 - the work has been composed by the student and describes his/her **own** work
 - the work has **not** been accepted in any previous application for a degree
 - all **verbatim extracts are distinguished by quotation marks**
 - all **sources of information are specifically acknowledged**
- acknowledgements
- an abstract (single line spaced) which summaries the key issues of the project
- a table of contents which specifies the main sections in the volume and the pages in which these start;
- a number of chapters which explain the work carried out and the context in which it developed.

14.1 Report Front Cover

This must be completed and signed by the student. **The work will not be accepted for assessment unless the front cover and its declaration has been included and signed.**

14.2 Title Page

MSc Project Report

Application of XML tagging
to the retrieval of sunbeams from cucumbers

Tommy Atkins

2018

A report submitted as part of the requirements for the degree of
MSc Big Data Analytics at Robert Gordon University, Aberdeen, Scotland

14.3 Declaration

A form declaring the original authorship of the content of the report should follow the title page.

Declaration

I confirm that the work contained in this MSc project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

Signed **Date.....**

The declaration must be signed and dated by the student when each volume of the report is submitted. **The work will not be accepted for assessment unless the above declaration has been included and signed.**

14.4 Format of the report

Broad guidelines are that the project report between 100 and 110 pages, excluding appendices. Each volume must be formatted for A4 size paper, in one and a half spacing. 11pt or 12pt font size should be used. You can use single spacing and 10pt font for your appendices. All pages should be numbered. All reports must be submitted online to the appropriate Moodle dropbox, as explained in Section 11.

In the main body of the report the following are to be included; the precise contents will depend on the nature of the project. They could contain with the chapters of the report.

1. Introduction and motivation
2. A review of relevant literature - an overview of the key issues/techniques/systems that have been considered, and their relation to the problem and its solution.
3. Discussion and evaluation of software tools to be used.
4. Problems identified and proposed solution(s).
5. Specification of the problem to be solved in the course of the subsequent project module, including a clear statement of aims and objectives. - Creating the Project Specification is an important task for the student, as it will form the basis of the work done. The description of a project as it appears in the portfolio is typically short and lacking in detail. Most descriptions include a statement of aims and objectives; some do not. Even when the original description does contain some statement of aims and objectives, the student will need to describe in detail what each of the aims and

objectives entails and to explain fully the motivation behind the project. This will typically involve analysis of alternatives and different interpretations of the objectives. Simply reproducing the original project description will not be sufficient.

6. A list of functional and non-functional requirements - The project report must include a list of functional and non-functional requirements expected of the software developed in the project.
7. Approach / Methodology adopted
8. Design of the solution
9. Implementation details of the developed solution including resources and facilities used. This may be elaborated into separate discussion of the overall system architecture, followed by details of individual software components.
10. Critical comparison between the original objectives and what has been achieved.
11. Critical evaluation of the student's work covering issues of efficacy, performance, efficiency, cost benefit and comparisons with other documented solutions.
12. A review of professional, social, legal and ethical issues with respect to the project.
13. Conclusions, which should include ways in which the work could have been improved and considerations for further developments.
14. A list of references (see section 14 and Appendix B below)
15. Appendices containing any useful material which is not in the body of the report. The following appendices should always be included:
 - Project plan (see below for details).
 - Detailed design documentation.
 - Detailed test data and results.
 - User guide.
 - Installation guide.

It is *not* necessary to include full code listings in the report, since the code will be included on the data-stick or CD.

The **project plan** should appear as an appendix to the project. The schedule should be presented in diagrammatic form, possibly using a tool like Microsoft Project; a description and rationale for each of the main activities; a brief review of required project resources, their availability and implications for the plan. As a guide, the project plan should comprise ~1000 words, i.e. around 3 pages.

Project Slides: The student may either create a physical project poster by printing the project slides, or may simply display the slides on a PC during the demonstration. In either case, the display should contain at most 10 to 12 slides. The slides will normally be prepared using Microsoft PowerPoint.

The project slides will be used for the purposes of illustration during the project demonstration.

15 Citation and References

The main text of each volume **must cite every piece of work whose ideas have been QUOTED or USED**. Refer to appendix ‘How to Avoid an Accusation of Plagiarism’ for more a more detailed discussion of why and how all written work must properly acknowledge the source of text and ideas borrowed from other authors.

Failure to observe the guidelines in Appendix ‘How to Avoid an Accusation of Plagiarism’ will lead to a charge of academic misconduct.

Note that it is permissible to include software (source code or executable binaries) produced by any other author in the implementation of the project solution. But such inclusion is subject to the conditions that

- the original author of the code is clearly identified, the extent of the “extracted” code is delimited (e.g. project source listings should identify the start and finish of the other author’s source code)
- the extracted code should constitute a minor part of the overall project implementation.

Failure to declare the original authorship of any part of the project code will lead to a charge of academic misconduct.

Note: Students are advised to assemble a correctly formatted bibliography and set of citations as they identify relevant sources during the course of the project. **Students should not leave this task to the last moment.** Unexpected events such as sickness or accident will not be accepted as an excuse for submitting a report that includes inadequately referenced sources.

Appendix A How to Avoid an Accusation of Plagiarism

Extracted from 'Guidelines for Literature Survey and Project Reports MSc/PgD Computing: Information Engineering January 2000'

It is vital that all written work (including research skills courseworks, seminar reports, workshop reports, literature survey and project report) contain the author's thoughts and text, and that all usage of text or ideas from anyone else is duly recognised by being acknowledged. Failure to do this leads to an accusation of plagiarism. The University's regulation on Academic Misconduct (Academic Regulation A3, 2012, Section 2, paragraph 6.1(i)) defines plagiarism as:

“**Plagiarism** is the practice of presenting the thoughts or writings of another or others as original, without acknowledgment of their source(s).”

Plagiarism is an example of academic misconduct and is treated extremely seriously by the University. A serious case of misconduct may even result in the student being forced to withdraw from the course.

More generally, plagiarism is also unacceptable. Equally serious sanctions are imposed on the rare occasion when plagiarism occurs in published books, conference proceedings, journal papers etc.

The definition of plagiarism does not refer to any particular source. The text or thoughts that are copied can come from a book, a journal or conference paper, a web page, a video, etc. There are occasions when it is very appropriate to quote exactly the text or thoughts of others; e.g. the definition of plagiarism above. This is **not** plagiarism since the source has been acknowledged and the copied text has been enclosed in quotation marks. This is an example of correctly and properly attributed text. In many cases it is also appropriate to name the author(s). The advice from Sloman (1992, page 16) is clear:

“If you include a quotation from another author make sure you give a FULL reference including page number so that it can be found quickly and easily. E.g., the reader may wish to check the context to see whether you have misunderstood.”

Again notice that this text has been correctly attributed. Notice however that such direct quotations tend to be quite short and there is often a good reason why the particular words should be used; e.g., the exact definition of plagiarism from the regulations, the very appropriate advice of Sloman.

Remember that a literature survey is supposed to show its author's understanding of the topic studied and its author's opinion of the contribution of the papers surveyed. Similarly, a project report often includes a critical appraisal of other people's work in comparison with the work of the author of the project report. In each case all relevant work must be appropriately acknowledged in citations and all copied text must be attributed to a particular author and paper and the text enclosed in quotation marks. Note that, even if you do not copy text from a particular paper, all text that describes or discusses the work or ideas of others should contain explicit citations attributing the work to particular people and publications.

As for the particular style to be used when citing references, there are numerous different citation styles in current use. For all student work, the Robert Gordon University requires the use of either Harvard or Vancouver style, which can be briefly characterised as follows. You are advised to **use Harvard style**. In Harvard style citations contain the author's surname and date. (The two citations in this appendix conform to Harvard style). In Vancouver style, citations just contain a number. Detailed descriptions of both styles may be found on the RGU library web pages; in addition, printed A4 leaflets may be obtained from the library. Links to electronic copies of these documents are provided on the Project Web Page.

Some software tools exist to help you manage your bibliography. For users of Microsoft Word, the RGU library has recently acquired the tool *Refworks* (see link on the Project Web Pages). For users of the LaTeX document formatting software, a similar function is performed by BibTeX.